Project 3 analysis

Knowledge based systems

Library

```
library(R.ROSETTA)
library(rJava)
```

```
Warning: package 'rJava' was built under R version 4.5.2
```

```
library(rmcfs)
```

```
Warning: package 'rmcfs' was built under R version 4.5.2
```

```

  ########################
  # rmcfs version 1.3.6 #
  ########################
  If used please cite the following paper:
  M. Draminski, J. Koronacki (2018),
  rmcfs: An R Package for Monte Carlo Feature Selection and Interdependency
Discovery,
  Journal of Statistical Software, vol 85(12), 1-28, doi:10.18637/
jss.v085.i12.
```

```
library(RCy3)
```

```
Warning: package 'RCy3' was built under R version 4.5.2
```

```
library(csVisuNet)
```

```
Loading required package: visNetwork
```

```
Loading required package: shiny
```

```
Warning: package 'shiny' was built under R version 4.5.2
```

```
Loading required package: shinydashboard
```

```
Warning: package 'shinydashboard' was built under R version 4.5.2
```

```

Attaching package: 'shinydashboard'
```

```
The following object is masked from 'package:graphics':

    box
```

```
Loading required package: shinythemes
```

```
Warning: package 'shinythemes' was built under R version 4.5.2
```

```
Loading required package: shinyjs
```

```
Warning: package 'shinyjs' was built under R version 4.5.2
```

```

Attaching package: 'shinyjs'
```

```
The following object is masked from 'package:shiny':

    runExample
```

```
The following object is masked from 'package:rJava':

    show
```

```
The following objects are masked from 'package:methods':

    removeClass, show
```

```
Loading required package: plyr
```

```
Loading required package: dashboardthemes
```

```
WARNING: the 'dashboardthemes' package is no longer being maintained as of
version 1.1.6, and will eventually be removed from CRAN. It is recommended
that users switch over to the 'bslib' or 'fresh' theming packages available on
CRAN.
```

```
Loading required package: tidyverse
```

```
Warning: package 'tibble' was built under R version 4.5.2
```

```
Warning: package 'tidyr' was built under R version 4.5.2
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
Warning: package 'purrr' was built under R version 4.5.2
```

```
Warning: package 'dplyr' was built under R version 4.5.2
```

```
Warning: package 'stringr' was built under R version 4.5.2
```

```
Warning: package 'lubridate' was built under R version 4.5.2
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.2.0     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.2     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.1
```

```
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::arrange()   masks plyr::arrange()
✖ purrr::compact()   masks plyr::compact()
```

```
✖ dplyr::count()     masks plyr::count()
✖ dplyr::desc()      masks plyr::desc()
✖ dplyr::filter()    masks stats::filter()
✖ dplyr::lag()       masks stats::lag()
✖ dplyr::mutate()    masks plyr::mutate()
✖ dplyr::rename()    masks plyr::rename()
✖ lubridate::show()  masks shinyjs::show(), rJava::show(), methods::show()
✖ dplyr::summarise() masks plyr::summarise()
✖ dplyr::summarize() masks plyr::summarize()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
Loading required package: backports
```

```
# library(vsn)
library(org.Mmu.eg.db)
```

```
Loading required package: AnnotationDbi
```

```
Warning: package 'AnnotationDbi' was built under R version 4.5.2
```

```
Loading required package: stats4
Loading required package: BiocGenerics
```

```
Warning: package 'BiocGenerics' was built under R version 4.5.2
```

```
Loading required package: generics

Attaching package: 'generics'

The following object is masked from 'package:lubridate':

    as.difftime

The following object is masked from 'package:dplyr':

    explain

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union
```

```
Attaching package: 'BiocGenerics'

The following object is masked from 'package:dplyr':

    combine

The following objects are masked from 'package:rJava':

    anyDuplicated, duplicated, unique

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min

Loading required package: Biobase
```

```
Warning: package 'Biobase' was built under R version 4.5.2
```

```
Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.

Loading required package: IRanges
```

```
Warning: package 'IRanges' was built under R version 4.5.2
```

```
Loading required package: S4Vectors
```

```
Warning: package 'S4Vectors' was built under R version 4.5.2
```

```


Attaching package: 'S4Vectors'
```

```
The following objects are masked from 'package:lubridate':

    second, second<-

The following objects are masked from 'package:dplyr':

    first, rename

The following object is masked from 'package:tidyr':

    expand

The following object is masked from 'package:plyr':

    rename

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname


Attaching package: 'IRanges'

The following object is masked from 'package:lubridate':

    %within%

The following objects are masked from 'package:dplyr':

    collapse, desc, slice

The following object is masked from 'package:purrr':

    reduce

The following object is masked from 'package:plyr':

    desc

The following object is masked from 'package:grDevices':

    windows
```

```
Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

    select
```

```
library(clusterProfiler)
```

```
Warning: package 'clusterProfiler' was built under R version 4.5.2
```

```
clusterProfiler v4.18.4 Learn more at https://yulab-smu.top/contribution-
knowledge-mining/

Please cite:

Guangchuang Yu, Li-Gen Wang, Yanyan Han and Qing-Yu He.
clusterProfiler: an R package for comparing biological themes among
gene clusters. OMICS: A Journal of Integrative Biology. 2012,
16(5):284-287

Attaching package: 'clusterProfiler'

The following object is masked from 'package:AnnotationDbi':

    select

The following object is masked from 'package:IRanges':

    slice

The following object is masked from 'package:S4Vectors':

    rename

The following object is masked from 'package:purrr':

    simplify

The following objects are masked from 'package:plyr':

    arrange, mutate, rename, summarise

The following object is masked from 'package:stats':
```

```
    filter
```

Data

Load the data.

```
df <- read.csv("Project3.csv")
```

```
df[, 65:71] |>
  head() |>
  knitr::kable()
```

SPEF2	GUSBA	CD300LD	TMEM40	F5	TTC23L	protectionStatus
−0.7391027	5.551729	1.3231816	1.513284	5.545211	1.4402210	NonProt
−2.9451678	5.668806	0.7552719	1.872456	6.225258	−0.4091149	Prot
−4.6397366	6.243670	−0.1161747	1.735303	5.777061	−1.1803050	Prot
−4.7306924	5.840112	1.9275191	2.600224	5.960179	0.3137017	NonProt
−3.2316905	6.764077	0.8557724	2.683193	6.431868	−1.1162132	Prot
−3.2041049	5.963313	1.6866660	3.395808	7.513857	−1.6191424	NonProt

```
table(df$protectionStatus) |>
  as.data.frame() |>
  `names<-`(c("Protection status", "Frequency")) |>
  knitr::kable()
```

Protection status	Frequency
NonProt	19
Prot	20

Monte Carlo Feature Selection

MCFS feature selection.

```
result_mcfs <- mcfs(
  protectionStatus ~ .,
  df,
  projections = 3000,
  projectionSize = 0.05,
  splits = 3,
  splitSetSize = 50,
```

```
  cutoffPermutations = 100,
  threadsNumber = 7
)
```

```
Checking input data...
Exporting params...
Exporting input data...
Running MCFS-ID...
Reading results...
Done.
```

```
plot(result_mcfs, type = "distances")
```

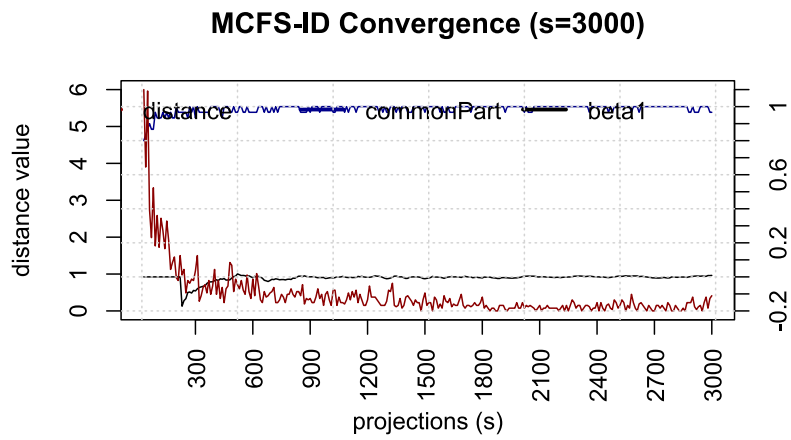


Figure 1: MCFS projection convergence.

Significant features.

```
result_mcfs$RI[1:result_mcfs$cutoff_value, ] |>
  knitr::kable()
```

	position	attribute	projections	classifiers	nodes	RI
3	1	X00000037843	170	0.8941176	0.9588235	0.1684384
35	2	SDK2	165	0.8565656	1.2303030	0.1615709
59	3	RWDD2A	199	0.7956449	0.9782245	0.1231151
61	4	LAD1	181	0.7845304	0.8839779	0.1132587
56	5	CDH20	178	0.7958801	0.9325843	0.1025256

	position	attribute	projections	classifiers	nodes	RI
69	6	F5	185	0.6630630	0.9171171	0.0951057
2	7	X00000053274	171	0.7660819	0.8596491	0.0941764

```
features <- result_mcfs$RI[result_mcfs$RI$RI > 0.05, ]$attribute
df_MCFS <- df[c(features, "protectionStatus")]
```

```
gid <- build.idgraph(result_mcfs, size = 20)
```

```
Selected 20 nodes and 60 edges.
```

```
plot.idgraph(gid, label_dist = 0.3)
```

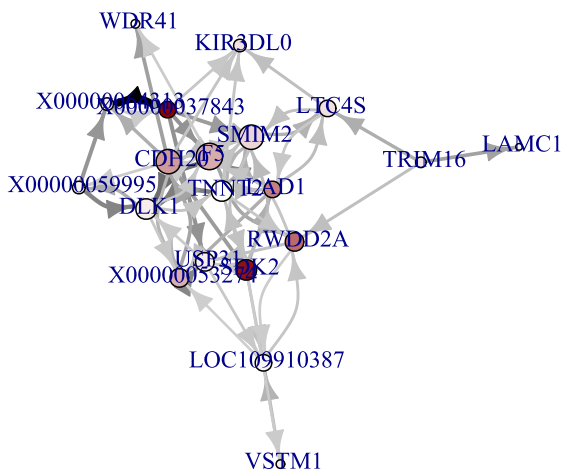


Figure 2: Interdependency graph.

ROSETTA

Can't do VSN normalization because only 39 rows. Could transpose, normalize, then transpose back?

```
JohnsonParam <- list(
  Module = TRUE,
  BRT = TRUE,
  BRTprec = 0.99,
  Precompute = TRUE,
  Approximate = TRUE,
  Fraction = 0.99
```

```
)

# df_vsn_full <- vsn::justvsn(as.matrix(df[, 1:70]))
# df_vsn_MCFS <- df_vsn_full[c(features, "protectionStatus")]

df_ros_list <- list(
  full = df,
  MCFS = df_MCFS
  #   full_vsn = df_vsn_full,
  #   MCFS_vsn = df_vsn_MCFS
)
```

Run rosetta on the full and MCFS datasets.

```
ros_full <- rosetta(
  df,
  roc = TRUE,
  clroc = "Prot",
  cvNum = 5,
  JohnsonParam = JohnsonParam
)
ros_MCFS <- rosetta(
  df_MCFS,
  clroc = "Prot",
  roc = TRUE,
  cvNum = 5,
  JohnsonParam = JohnsonParam
)
```

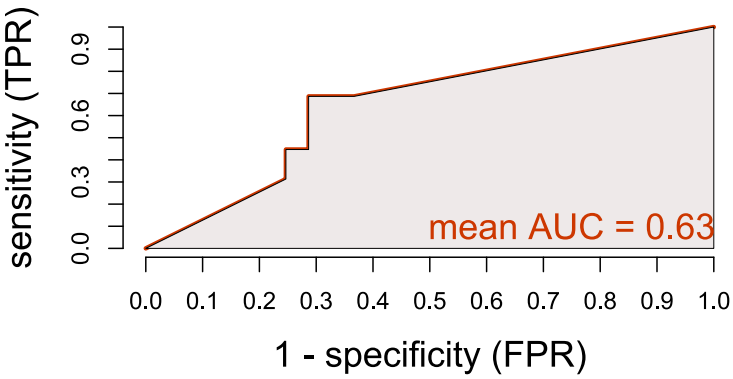
QC rosetta on full dataset.

```
ros_full$quality |>
  tidyr::pivot_longer(
    cols = everything(),
    names_to = "metric",
    values_to = "value"
  ) |>
  knitr::kable()
```

metric	value
accuracyMean	0.585714
accuracyMedian	0.625000
accuracyStd	0.087846
accuracyMin	0.428571

metric	value
accuracyMax	0.625000
ROC.AUC	0.650000
ROC.AUC.SE	0.253962
ROC.AUC.MEAN	0.628333
ROC.AUC.MEDIAN	0.600000
ROC.AUC.STDEV	0.078085
ROC.AUC.MIN	0.541667
ROC.AUC.MAX	0.750000
ROC.AUC.SE.MEAN	0.224116
ROC.AUC.SE.MEDIAN	0.221682
ROC.AUC.SE.STDEV	0.027703
ROC.AUC.SE.MIN	0.183894
ROC.AUC.SE.MAX	0.253962

```
plotMeanROC(ros_full)
```



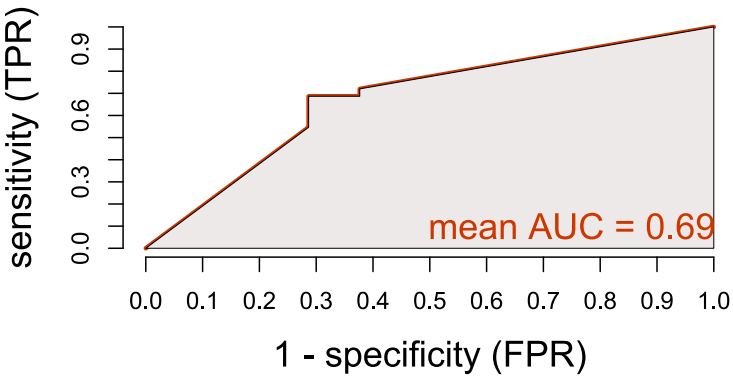
QC on rosetta MCFS dataset.

```
ros_MCFS$quality |>
  tidyr::pivot_longer(
    cols = everything(),
    names_to = "metric",
```

```
    values_to = "value"
  ) |>
  knitr::kable()
```

metric	value
accuracyMean	0.621429
accuracyMedian	0.625000
accuracyStd	0.145839
accuracyMin	0.500000
accuracyMax	0.857143
ROC.AUC	1.000000
ROC.AUC.SE	0.000000
ROC.AUC.MEAN	0.689167
ROC.AUC.MEDIAN	0.625000
ROC.AUC.STDEV	0.182327
ROC.AUC.MIN	0.533333
ROC.AUC.MAX	1.000000
ROC.AUC.SE.MEAN	0.173906
ROC.AUC.SE.MEDIAN	0.215141
ROC.AUC.SE.STDEV	0.097996
ROC.AUC.SE.MIN	0.000000
ROC.AUC.SE.MAX	0.231683

```
plotMeanROC(ros_MCFS)
```



Rules

Look at rules for full dataset.

```
rules_full <- ros_full$main
rls_ros_full <- viewRules(
  rules_full[rules_full$decision == "Prot", ],
  setLabels = TRUE,
  labels = c("low", "medium", "high")
)
rls_ros_full[1:12, ] |> knitr::kable()
```

rule	length	accu-racy	sup-port	pValue
IF SDK2(high) THEN Prot	1	1	10	0.0148213
IF LOC109910387(low) THEN Prot	1	1	10	0.0148213
IF X00000053274(high) AND X00000059200(high) THEN Prot	2	1	6	0.6058806
IF FABP5(medium) AND X00000053012(high) THEN Prot	2	1	6	0.6058806
IF X00000059200(high) AND SDK2(high) THEN Prot	2	1	6	0.6058806
IF FABP5(low) AND VSTM1(low) THEN Prot	2	1	4	1.0000000
IF X00000053274(medium) AND VSTM1(low) THEN Prot	2	1	4	1.0000000
IF FABP5(medium) AND LILRB1(low) THEN Prot	2	1	3	1.0000000
IF LOC719667(low) AND KIR3DL0(high) THEN Prot	2	1	4	1.0000000
IF LOC719667(medium) AND LILRB1(low) THEN Prot	2	1	5	1.0000000

rule	length	accuracy	support	pValue
IF FABP5(low) AND X00000053274(medium) THEN Prot	2	1	4	1.0000000
IF X00000053274(medium) AND C5H4orf45(low) THEN Prot	2	1	4	1.0000000

Look at MCFS rules.

```
rules_mcfs <- ros_MCFS$main
rls_ros_MCFS <- viewRules(
  rules_mcfs[rules_mcfs$decision == "Prot", ],
  setLabels = TRUE,
  labels = c("low", "medium", "high")
)
rls_ros_MCFS[1:12, ] |> knitr::kable()
```

rule	length	accuracy	support	pValue
IF SDK2(high) THEN Prot	1	1	10	0.0174368
IF LOC109910387(low) THEN Prot	1	1	10	0.0174368
IF X00000037843(low) AND VSTM1(low) THEN Prot	2	1	6	0.7128007
IF X00000037843(low) AND RWDD2A(high) THEN Prot	2	1	6	0.7128007
IF SDK2(medium) AND WDR41(high) THEN Prot	2	1	3	1.0000000
IF SDK2(medium) AND LAD1(low) THEN Prot	2	1	1	1.0000000
IF X00000037843(low) AND RWDD2A(low) THEN Prot	2	1	2	1.0000000
IF SDK2(low) AND VSTM1(low) THEN Prot	2	1	1	1.0000000
IF X00000037843(medium) AND SMIM2(low) THEN Prot	2	1	4	1.0000000
IF X00000037843(medium) AND CDH20(high) THEN Prot	2	1	3	1.0000000
IF X00000037843(low) AND LAD1(high) THEN Prot	2	1	1	1.0000000
IF X00000037843(low) AND RWDD2A(medium) THEN Prot	2	1	3	1.0000000

Recalculate rules

Recalculate rules on full dataset.

```r
rec_rules_full <- recalculateRules(df, rules_full)
rec_rls_ros_full <- viewRules(
  rec_rules_full[rec_rules_full$decision == "Prot", ],
  setLabels = TRUE,
  labels = c("low", "medium", "high")
)
rec_rls_ros_full[1:12, ] |> knitr::kable()
```

rule	length	accu-racy	sup-port	pValue
IF LOC109910387(low) THEN Prot	1	1	14	0.0001310
IF SDK2(high) THEN Prot	1	1	11	0.0051108
IF X00000053274(high) AND X00000059200(high) THEN Prot	2	1	9	0.0404217
IF X00000059200(high) AND SDK2(high) THEN Prot	2	1	8	0.1044226
IF X00000053274(high) AND SDK2(high) THEN Prot	2	1	7	0.2570403
IF FABP5(medium) AND X00000053012(high) THEN Prot	2	1	6	0.6058806
IF FABP5(low) AND VSTM1(low) THEN Prot	2	1	6	0.6058806
IF X00000053274(medium) AND X00000037843(low) THEN Prot	2	1	6	0.6058806
IF FABP5(medium) AND X00000037843(low) THEN Prot	2	1	6	0.6058806
IF X00000053274(medium) AND VSTM1(low) THEN Prot	2	1	4	1.0000000
IF FABP5(medium) AND LILRB1(low) THEN Prot	2	1	4	1.0000000
IF LOC719667(low) AND KIR3DL0(high) THEN Prot	2	1	4	1.0000000

Recalculate rules on MCFS dataset.

```r
rec_rules_mcfs <- recalculateRules(df_MCFS, rules_mcfs)
rec_rls_ros_MCFS <- viewRules(
  rec_rules_mcfs[rec_rules_mcfs$decision == "Prot", ],
  setLabels = TRUE,
  labels = c("low", "medium", "high")
)
rec_rls_ros_MCFS[1:12, ] |> knitr::kable()
```

rule	length	accu-racy	sup-port	pValue
IF LOC109910387(low) THEN Prot	1	1	14	0.0001542

rule	length	accu-racy	sup-port	pValue
IF SDK2(high) THEN Prot	1	1	11	0.0060127
IF X00000037843(low) AND VSTM1(low) THEN Prot	2	1	7	0.3024003
IF X00000037843(low) AND LAD1(medium) THEN Prot	2	1	7	0.3024003
IF X00000037843(low) AND RWDD2A(high) THEN Prot	2	1	6	0.7128007
IF X00000037843(medium) AND MYH15(high) THEN Prot	2	1	6	0.7128007
IF X00000037843(low) AND F5(medium) THEN Prot	2	1	6	0.7128007
IF RWDD2A(medium) AND CDH20(high) THEN Prot	2	1	6	0.7128007
IF X00000037843(low) AND WDR41(high) THEN Prot	2	1	6	0.7128007
IF SDK2(high) AND LAD1(medium) THEN Prot	2	1	6	0.7128007
IF SDK2(medium) AND WDR41(high) THEN Prot	2	1	4	1.0000000
IF SDK2(medium) AND LAD1(low) THEN Prot	2	1	2	1.0000000

VisuNet

Error in `colnames<-`: ! attempt to set 'colnames' on an object with less than two dimensions

```
cytoscapePing()
visunetcyto(ros_full$main) # Failed to load into cytoscape
vis_full <- visunet(ros_full$main)
saveRDS(vis_full, "vis_full.RDS")
```

But this worked.

```
cytoscapePing()
visunetcyto(ros_MCFS$main)
vis_mfcs <- visunet(ros_MCFS$main)
saveRDS(vis_mfcs, "vis_mcfs.RDS")
```

Interdependency graph for MCFS dataset.

```
gid <- build.idgraph(result_mcfs, size = 20)
```

```
Selected 20 nodes and 60 edges.
```

```
plot.idgraph(gid, label_dist = 0.3)
```

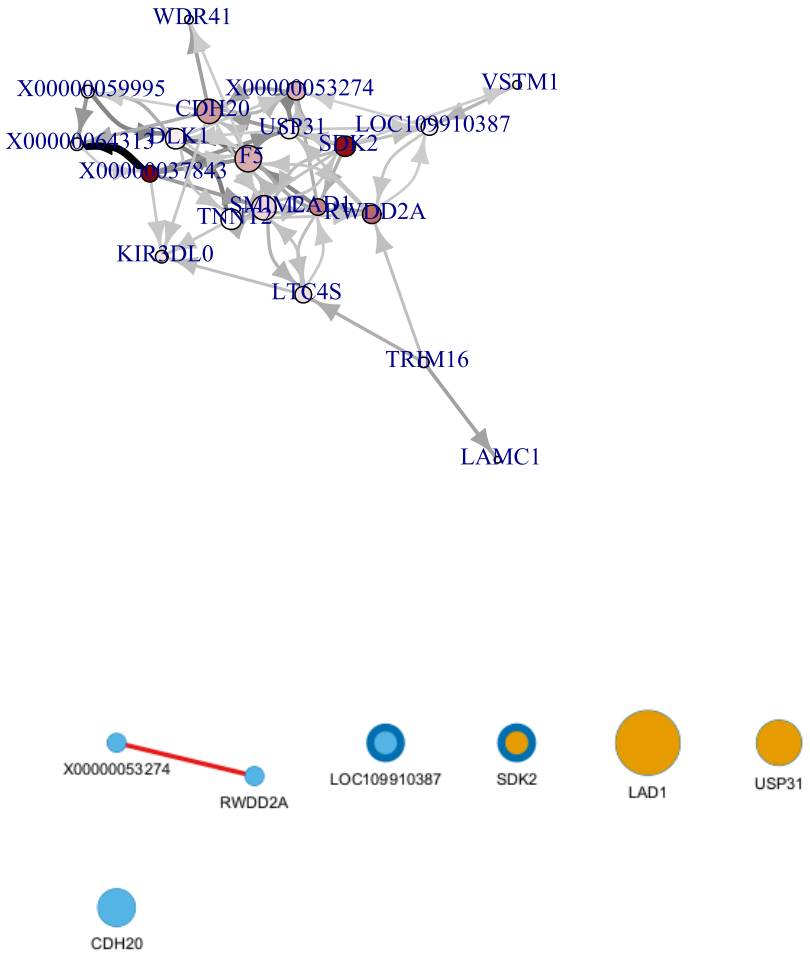


Figure 3: Network MCFS all

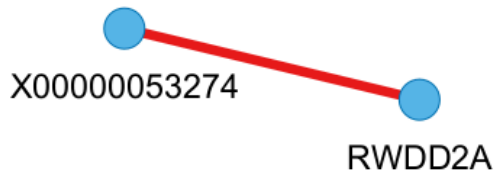


Figure 4: Network MCFS Non-protection

Rule visualization

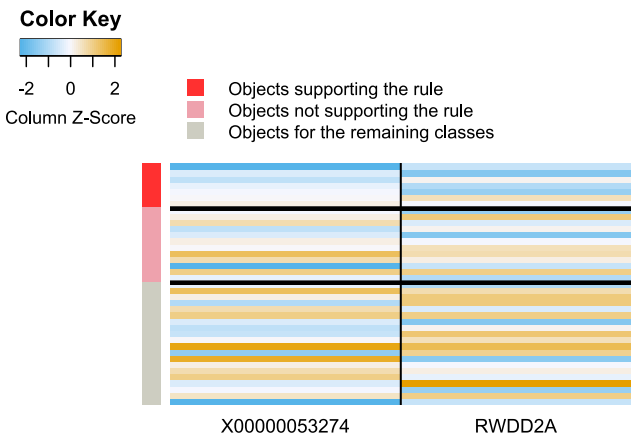
Visualize the new rules on the full dataset. This fails.

```
# plotRule(
#   df,
#   rec_rules_full,
#   type = "heatmap",
#   discrete = FALSE,
#   ind = 5,
#   label = c('low', 'medium')
# )
# plotRule(
#   df,
#   rec_rules_full,
#   type = "boxplot",
#   discrete = FALSE,
#   label = c('low', 'medium')
# )
```

Visualize the new rules on the MCFS dataset.

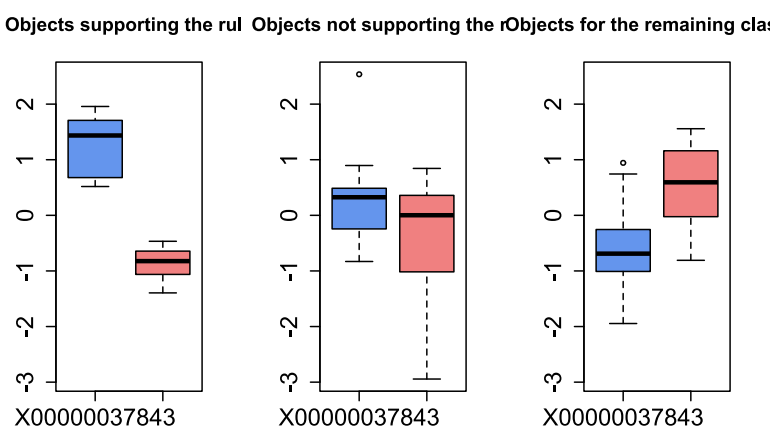
```
# set new labels for plots using parameter label=c()
plotRule(
  df_MCFS, # dataset
  rec_rules_mcfs, # rules
  type = "heatmap",
  discrete = FALSE,
  ind = 5,
```

```
  label = c('low', 'medium')
)
```



ns IF RWDD2A=low AND X00000053274=medium THEN NonP

```
plotRule(
  df_MCFS,
  rec_rules_mcfs,
  type = "boxplot",
  discrete = FALSE,
  label = c('low', 'medium')
)
```



s IF X00000037843=low AND CDH20=medium THEN NonPrc

Pathway analysis

Load the visuNet results back in.

```r
vis_full <- readRDS("vis_full.RDS")
vis_mcfs <- readRDS("vis_mcfs.RDS")
```

How to do pathway analysis?

```r
gene_entrez_full <- bitr(
  unique(as.character(vis_full$all$nodes$label)),
  fromType = 'SYMBOL',
  toType = c("ENTREZID"),
  org.Mmu.eg.db
)
```

```
'select()' returned 1:1 mapping between keys and columns
```

```r
gene_entrez_mcfs <- bitr(
  unique(as.character(vis_mcfs$all$nodes$label)),
  fromType = 'SYMBOL',
  toType = c("ENTREZID"),
  org.Mmu.eg.db
)
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Warning in bitr(unique(as.character(vis_mcfs$all$nodes$label)), fromType =
"SYMBOL", : 14.29% of input gene IDs are fail to map...
```

```r
genes_GO_full <- groupGO(
  gene = unique(gene_entrez_full$ENTREZID),
  OrgDb = org.Mmu.eg.db,
  ont = "MF",
  level = 5,
  readable = TRUE
)

genes_GO_mcfs <- groupGO(
  gene = unique(gene_entrez_mcfs$ENTREZID),
  OrgDb = org.Mmu.eg.db,
  ont = "MF",
  level = 5,
  readable = TRUE
)
```

Plots fail.

```
# x1 <- summary(genes_GO_full)[
#   which(summary(genes_GO_full)$Count > 1),
#   c('ID', 'Count')
# ]
# x <- as.numeric(x1[, 2])
# names(x) <- x1$ID
# barplot(x, col = rainbow(20), las = 2)
```

```
# x1 <- summary(genes_GO_mcfs)[
#   which(summary(genes_GO_mcfs)$Count > 1),
#   c('ID', 'Count')
# ]
# x <- as.numeric(x1[, 2])
# names(x) <- x1$ID
# barplot(x, col = rainbow(20), las = 2)
```

```
genes_GO_full@result[genes_GO_full@result$Count > 0, ] |>
  knitr::kable()
```

ID	Description	Count	GeneRatio	geneID
GO:0008234	GO:0008234 cysteine-type peptidase activity	1	1/2	USP31
GO:0019783	GO:0019783 ubiquitin-like protein peptidase activity	1	1/2	USP31

```
genes_GO_mcfs@result[genes_GO_mcfs@result$Count > 0, ] |>
  knitr::kable()
```

ID	Description	Count	GeneRatio	geneID
GO:0008234	GO:0008234 cysteine-type peptidase activity	1	1/6	USP31
GO:0019783	GO:0019783 ubiquitin-like protein peptidase activity	1	1/6	USP31
GO:0045296	GO:0045296 cadherin binding	1	1/6	CDH20
GO:0043169	GO:0043169 cation binding	1	1/6	CDH20
GO:0004896	GO:0004896 cytokine receptor activity	1	1/6	SDK2